Help Desk: Final Shift

A CTF for Cisco NetAcad **Hardware and Upgrade Support** (Modules 1–2) · Game 4 · Facilitator guide & answer key · **Instructor only**

> **At a glance:** 39 challenges · 7,700 points · 9 categories (all start with "UP") following topics 1.1–2.5 · 13 evidence files · loads next to Games 1–3 · tested on CTFd 3.8.6 alongside all three.

The story

In the last stop on the IT Support Specialist path, students become the **Hardware & Upgrade** technician. The work covers ESD and electrical safety; finding device information on Windows and Mac; identifying ports, cables, adapters and converters; Device Manager codes; and a firmware rollback decision. Then come compatibility planning, PSU connectors and sizing, CPU sockets, M.2 SATA vs NVMe, PCIe lanes, USB math, driver tools, and e-waste laws and disposal records. In the boss ticket, students pick the one RAM kit that matches the board's memory type, appears on the QVL, and gets the designer to 32 GB.

> **Built from your decks:** every answer cites its source slide ("Source: M# S##" = Module and slide number in the instructor PPTX).

Setup

1. Set up CTFd and finish the setup wizard. See `docs/platform-setup.md` in the OmniTech-Arena repository.
2. In CTFd as admin: **Settings** → **Access Tokens** → **Generate**.
3. From the repository root: `pip install requests`, then `python tools/deploy.py tracks/helpdesk-cisco-it-support/04-final-shift_hardware-and-upgrade-support --url http://<host>:8000 --token ctfd_...`
4. Look for `Done: 39 challenges loaded.` Run it **once**. Other games aren't touched. Add `--hidden` to load the game hidden.
5. One time per server: paste the two sort functions from `docs/platform-setup.md` into **Admin Panel** → **Config** → **Theme** so each game's categories stay grouped.

Running it alongside Canvas

- Each category maps to NetAcad topics, so students can do "UP 2B" right after finishing topic 2.2.
- Everything can be solved from the files and the course. Hands-on extension: have students check their answers to *Tool Time* and *Mac Equivalents* on real machines, and open a lab PC to physically identify the *Plug It In* connectors.
- Good debriefs: *Firmware Flashback* (why "newest safe" beats "oldest"), *Paper Trail Gap* (printers store data too), and the boss (why the QVL matters even when a kit is the right type).

Answer key

Every answer below was submitted by a test student on a CTFd 3.8.6 server that already had Games 1–3 loaded, and all 39 were accepted (full score: 7,700). Common wrong answers were checked and rejected, for example PC C, firmware 4.1.6, a 450 W PSU, and RAM-13 in the boss.

Challenge	Answer	Why / discussion	Files
UP 1A Safety & Device Info (1.1-1.2)			
Final Shift 50 pts	`final shift`	Source:	–
Feel the Zap 150 pts 3 tries	`3000,30`	Commas are optional (3000 or 3,000). Source: M1 S20 ESD	–
Safe or Sorry? 200 pts 3 tries	`ACDF`	Source: M1 S19-20 electrical safety and ESD	–
Tool Time 250 pts 4 tries	`CDAB`	Source: M1 S25-32 Windows tools	–
Run Box Shortcut 100 pts 3 tries	`msinfo32`	Source: M1 S28	–
Mac Address on a Mac 200 pts 3 tries	`a4:83:e7:2b:9c:51`	Source: M1 S42 ifconfig	ifconfig_MKT-MBP04.txt
Mac Equivalents 150 pts 3 tries	`BAC`	Source: M1 S36, S40, S42	–
Will It Run? 300 pts 2 tries	`B`	A fails RAM (8 GB). C has integrated graphics with shared memory, not a 4 GB discrete GPU. Only 2 attempts. Source: M1 S61-62 application compatibility	app_compatibility_check.txt
UP 1B Ports & Cables (1.3)			
Name That Port 250 pts 4 tries	`DAFCEB`	Source: M1 S45-50 legacy and video ports	–
Adapter or Converter? 200 pts 3 tries	`ACAAC`	Source: M1 S56-57 adapters and converters	–
Big Iron Power 150 pts 3 tries	`C19`	Source: M1 S54 power cables	–
Thunder & Type-C 150 pts 3 tries	`Thunderbolt`	Source: M1 S51, S53	–
UP 1C Hardware Issues (1.4)			
Dead PC Checklist 150 pts 2 tries	`BCA`	Source: M1 S60 'Troubleshooting Power Issues'	–
Down Arrow 250 pts 3 tries	`Realtek PCIe GbE Family Controller`	Code 22 = disabled. The printer is Code 43 (error), and the fingerprint reader is Code 28 (no driver). Source: M1 S64 / M2 S62 Device Manager status icons	device_manager_ACC-PC11.txt
Firmware Flashback 300 pts 2 tries	`4.2.0`	4.2.1 has the failover bug this router hits. 4.2.0's issue only affects more than 250 clients, so it's the newest safe version. 4.1.6 works but isn't the newest safe one. Only 2 attempts. Source: M1 S65 'Firmware Update Issues'	router_firmware_notes.txt
Did You Reboot? 100 pts 3 tries	`power cycling`	Source: M1 S59 'Did You Reboot?'	–

UP 2A Architecture & Compatibility (2.1)			
Software Layers 200 pts 3 tries	`CDAB`	Source: M2 S23 The Role of Software	–
Should Have Planned 250 pts 3 tries	`BCDA`	Source: M2 S24-25 planning the installation or upgrade	–
Tested and Approved 100 pts 3 tries	`QVL`	Source: M2 S26 HCL/QVL	–
UP 2B Core Components (2.2)			
Plug It In 250 pts 4 tries	`DFABEC`	Source: M2 S32-33 power supply connectors	–
Twelve Volt Club 100 pts 2 tries	`12`	Source: M2 S34 power supply voltage	–
Socket Match 300 pts 3 tries	`2341`	CPU-1 LGA1700 to MB-2, CPU-2 AM5 to MB-3, CPU-3 LGA1151 to MB-4, CPU-4 AM4 to MB-1. Source: M2 S40-41 CPU socket types	parts_bin.csv
Small Box 150 pts 3 tries	`Mini-ITX`	Source: M2 S38 motherboard form factors	–
Wrong Slot 250 pts 2 tries	`M2_2`	M2_1 is NVMe-only. M2_2 accepts SATA M.2. Source: M2 S45-46 SATA, NVMe, M.2	motherboard_manual_storage.txt
North or South? 150 pts 2 tries	`Southbridge`	Source: M2 S37 motherboard chipset	–
Forget Me Not 100 pts 1 tries	`RAM`	Source: M2 S43 types of memory	–
UP 2C Interfaces & Expansion (2.3)			
Lanes for Cards 200 pts 3 tries	`PCI_E1,PCI_E2`	Source: M2 S49-50 PCIe lane sizes	pcie_slot_layout.txt
Speed of a Flash Drive 250 pts 4 tries	`500`	30 x 8,000 / 480 = 500 s (about 8.3 minutes). Source: M2 S52 USB versions table	–
Size the PSU 300 pts 4 tries	`382,550`	Sum = 382 W. x 1.3 = 496.6 W, so the smallest PSU that covers it is 550 W. Source: M2 S25 calculate power needs, S54 GPU power	psu_sizing_worksheet.txt
Wireless, Two Ways 150 pts 2 tries	`UIU`	Source: M2 S55-56 wireless and Bluetooth cards	–
Integrated vs Discrete 150 pts 2 tries	`integrated`	Source: M2 S53-54 GPUs	–
UP 2D Drivers & Device Manager (2.4)			
Driver Tab Decisions 250 pts 4 tries	`CAEDB`	Source: M2 S65 Driver tab options	–
Printer Fix 150 pts 2 tries	`Roll Back Driver`	Source: M1 S63, M2 S65	device_manager_ACC-PC11.txt
WHQL? 150 pts 3 tries	`Windows Hardware Quality Labs`	Source: M2 S59	–

UP 2E E-Waste & Disposal (2.5)			
Which Law? 250 pts 3 tries	`BADCE`	Source: M2 S69-71 data security and regulations	–
Certified Recyclers 150 pts 3 tries	`e-Stewards`	Source: M2 S72	–
Paper Trail Gap 300 pts 3 tries	`PRN-HR-01`	The multifunction printer had a 320 GB HDD and NONE RECORDED. The monitor and keyboard have no storage. Source: M2 S69, S73	asset_disposal_log.csv
Toxic Trio 100 pts 2 tries	`cadmium`	Source: M2 S70	–
UP 3 Boss Ticket			
The 32 GB Upgrade 500 pts 3 tries unlocks after: Will It Run?, Socket Match, Tested and Approved	`RAM-12`	Board = MSI B760M (DDR4), so DDR5 RAM-11 is out. RAM-13 isn't on the QVL. RAM-14 is only 16 GB. RAM-15 (2x32 GB) isn't tested. That leaves RAM-12. Unlocks after Will It Run?, Socket Match, and Tested and Approved. Source: M1 S61-62 compatibility, M2 S24-26 compatibility/QVL, S43 RAM	boss_ticket_INC-30418.txt msinfo32_DSN-PC02.txt ram_inventory.csv qvl_MSI_B760M_DDR4.txt